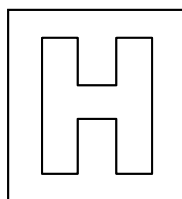


Candidate Name: _____

Class Adm No

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2013 Preliminary Examination 2

Pre-University 3

H2 Biology

9648/03

20 September 2013

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

For Examiner's Use	
1	
2	
3	
4	
5	
Total	

This question paper consists of 16 printed pages.

[Turn over

1. The pink bollworm moth, *Pectinophora gossypiella*, is a pest of cotton crops. The size of its population can be reduced by releasing large numbers of sterile male moths into cotton fields. The sterile male moths mate with wild females from the cotton fields, but no offspring are produced.

(a) Explain why, in gene technology:

- 1. easier to, identify / screen;
 - 2. more economical / time saving / labour saving / harmless;
 - 3. resistance gene(s) can be passed to other bacteria;
 - 4. idea of antibiotics no longer effective
or
requiring development of new antibiotics;

[2]

1. promoter, initiates transcription / switches on gene /causes gene expression / **AW**;
2. ref. binding of, RNA polymerase / transcription factors;
3. otherwise gene has to be inserted near an existing promoter;
4. this is difficult to do / this may disrupt expression of existing gene;
5. in eukaryotes precise position of promoter important;
6. idea that you need a coral promoter to switch on a coral gene;

- (i)** Outline the principles of gel electrophoresis.

- [4]

- [2]

- [2]

- (d) The United States Department of Agriculture has ruled that the release of sterile males to control insect pest numbers is environmentally preferable to all other alternatives.

Suggest what information would be needed to determine whether the release of the sterile male moths, carrying the gene for DsRed, has a damaging effect on the environment.

- | | | |
|------|--|-------|
| | 1. that DsRed is not toxic to predators of the moth; | |
| | 2. that DsRed does not persist in the environment; | |
| | 3. that the gene cannot pass to other organisms; | |
| | 4. does not alter, food web / ecosystem, (in harmful way); | |

.....

..... [2]

[Total: 15]

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2.

- (a) Plant responses to environmental changes are co-ordinated by plant growth substances (plant hormones).

Suggest why plants need to be able to respond to their environment.

to cope with changing conditions / AW ; R adapt	
avoid abiotic (non-living chemical and physical factors in the environment) stress ;	
to maximise photosynthesis	
or	
to obtain more, light / water / minerals ; ora	
avoid, herbivory / grazing ;	
(methods of preventing grazing could include producing more toxins / more spines / encouraging stinging ants) R predation	 [2]
to ensure, germination in suitable conditions / pollination /	
seed dispersal ;	

- each batch of seeds was placed on moist filter paper in a Petri dish and given a different treatment

The different treatments are shown in Table 2.1. Each tick represents one of the eight batches of seeds.

Table. 2.1

	Treatment	concentration of gibberellin (mol dm^{-3})			
		0.00	0.05	0.50	5.00
A	Water	✓	✓	✓	✓
B	Absciscic acid	✓	✓	✓	✓

The batches of seeds were left to germinate at 25°C in identical conditions and the percentage germination was calculated

Fig. 2.1 shows the results of this investigation

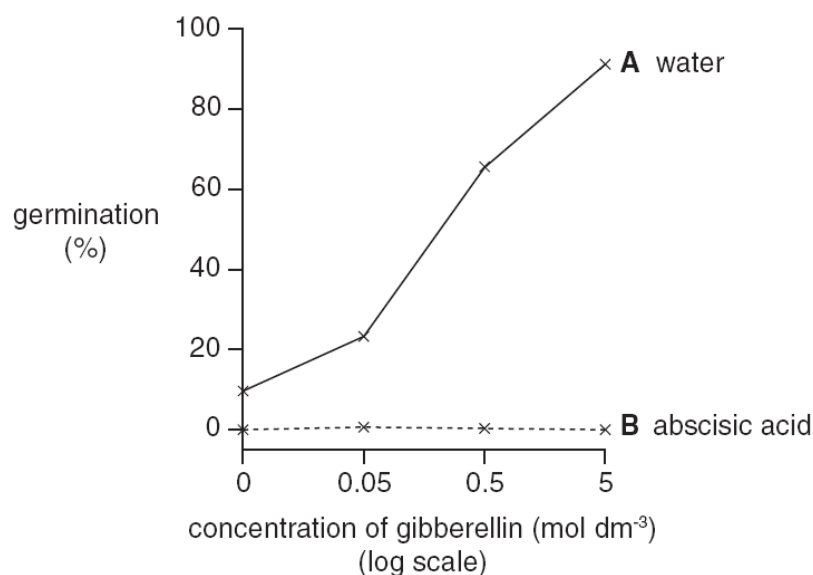


Fig. 2.1

- (i) Describe, with reference to Fig. 2.1, the effects of the plant growth substances on the germination of lettuce seeds.

..... in water / in A / with no abscisic acid,
germination increases as conc. GA increases ;

..... when abscisic acid present / in B, no germination ;

..... maximum germination 90% with 5 mol dm⁻³ GA,
in water / without abscisic acid ; A 91% (± 2%) for 90%

..... 2 comparative figures (x and y refs. plus units) ;

..... GA concentration increases,
logarithmically / by a factor of 10, on x axis ;

..... 10 times more GA gives,
3 (conc 0.05 to 0.5) / 0.5 (conc 0.5 to 5),
times more germination ;

..... [4]

- (ii) Explain why all the lettuce seeds were kept at 25 °C.

..... so temperature doesn't affect results /
so only desired variable(s) changed /
to show just the effect of plant hormones ; A fair test

..... since temperature affects enzyme activity / "optimum temperature for enzyme activity" /
"this is the temperature when enzymes work best";

..... suitable / optimum, temperature for (lettuce) germination ;

..... [2]

- (iii) State three factors that could affect the results of the experiment.

1 volumes of liquid(s) ;
ABA concentration ;
oxygen availability ;
age of seeds ;
previous storage of seeds / viability idea ;

2 genotype / variety, of seeds ;
size / type of, petri dish / filter paper ;
length of time experiment left for (before recording results) ;

3 space between seeds ;
light qualified (duration / intensity / wavelength);
use of distilled water;
all lids, off / on;

..... [3]

- (c) Propose two commercial uses of plant growth substances.

seedless, fruits / grapes ;
weedkillers ;
rooting powder / to grow cuttings / used in tissue culture ;
control fruit ripening ;
controls fruit drop ;
restrict hedge growth ;
preserve, cut flowers / green vegetables ;
specific example of improved fruit quality ;
producing malt / in brewing ;

..... [2]

[Total: 13]

3. Genetic information in humans can be obtained by DNA profiling.

In DNA profiling, the polymerase chain reaction is used by a scientist to amplify a particular sequence of DNA.

(a) Briefly describe the steps of this technique.

heat DNA to approximately 90 °C or to separate strands;
cool to approximately 50 °C or to anneal/attach primers;
heat to approx 72 °C or Taq/DNA polymerase copies strands;
repeat cycle;

..... [4]

Scientists investigating the performance of athletes found that one gene contributing to the performance of sprinters is the *ACTN3* gene. There are two alleles of the gene, the 577R allele and the 577X allele. The 577X allele codes for a very short protein fragment in muscle fibres due to a stop codon mutation.

Table 3.1 summarises the athletic potential for the three possible genotypes for the *ACTN3* gene.

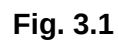
Table. 3.1

ACTN3 genotype	Athletic potential
577R / 577R	outstanding sprinter
577R / 577X	good sprinter or long-distance runner
577X / 577X	very good long-distance runner

A scientist tested sprinters to see if they possessed the 577R allele. Samples were obtained from athletes' muscle fibres. A standard containing proteins of the same lengths as the proteins coded for by both alleles 577X and 577R was used as a comparison. The standard and the samples were exposed to gel electrophoresis. In gel electrophoresis, protein molecules separate in the same way as DNA molecules.

Standard

Good
Sprinter



- Explain why you have placed the bands in these positions.

[4]

A cybrid (cytoplasmic hybrid cell) is produced as shown in Fig. 3.2.

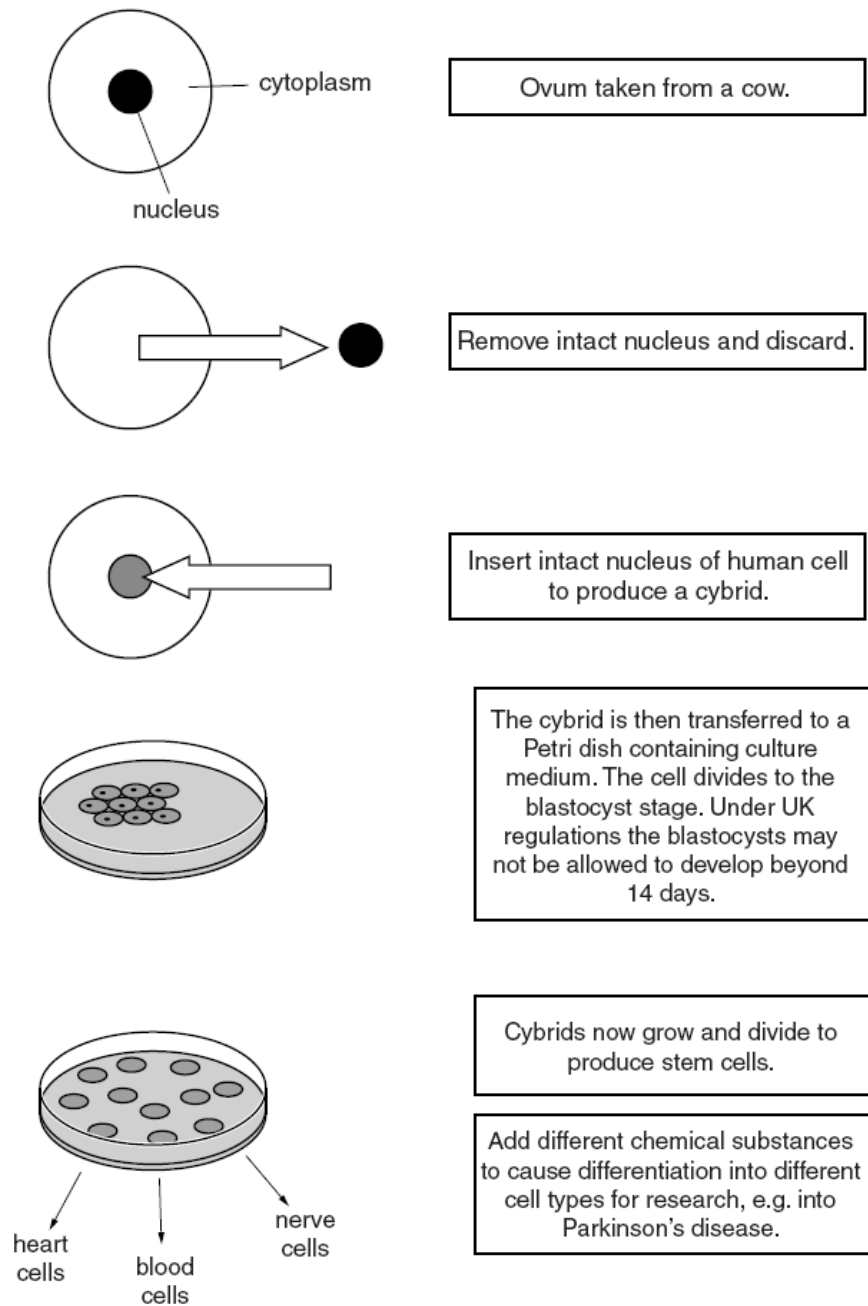


Fig. 3.2

- (c) The DNA of such a cybrid is 99.6% human. The remaining 0.4% of the DNA is in the cytoplasm. Explain why there is DNA in the cytoplasm of the cybrid.

.....	1 (mitochondrial DNA) from the cow / not of human origin ;	
.....		
.....		
.....		[1]

- (d) When the Human Fertilisation and Embryology Bill was considered by the UK Parliament in 2008, some people argued that it is unethical to allow the production of cybrids.

State whether you agree or disagree that this is unethical and explain why you reached this decision.

unethical:

- 1 presence of human and animal DNA means it is partly human and partly cow / a humananimal hybrid which would not happen in nature ;
- 2 claims of the benefits of embryonic stem cell research are over-rated / few (if any) examples of success in medical applications ;
- 3 dangerous precedent / thin end of the wedge argument / current applications (may be benign) but will lead to abuse in future ;
- 4 possibility of unforeseen (irreversible) consequences ;
- 5 unnecessary because, there already are / will soon be, alternative techniques (to the use of embryonic stem cells) ;
- 6 e.g. use of adult stem cells/umbilical cord cells ;

ethical:

- 1 the amount of non-human DNA is very small/negligible ;
- 2 protocol limits keeping 'embryo' to 14 days (to guard against abuse) so cannot develop into a whole organism ;
- 3 provides a more ethical alternative to the use of embryonic stem cells from humans ;
- 4 can provide many more stem cells than is possible from embryonic stem cells ;
- 5 it would be unethical not to use cybrids to relieve, human suffering / e.g. diseases such as Parkinson's disease ;
- 6 techniques which offer alternatives to embryonic stem cells builds on work done initially on embryonic stem cells ;
- 7 idea that ovum has bovine proteins which would be rejected at implantation ;

Allow other reasonable arguments on either side. If a candidate does not express an opinion, limit score to one mark unless an attempt has been made to argue that it is a finely balanced matter. Ignore references to 'playing God' and 'against religion'.

..... [3]

[Total: 12]

4. The enzyme urease is a catalyst of the hydrolysis of urea in solution, forming ammonia and carbon dioxide, for example in the breakdown of urea in soils by microorganisms.

You are required to plan an investigation to compare the activity of urease free in solution and urease immobilised in alginate beads.

As the reaction proceeds, the ammonia released dissolves, causing the pH to increase.

You are provided with the following equipment which you may use or not in your plan, as you wish. You may not use any additional equipment in your plan.

- an unlimited supply of calcium alginate beads all of uniform size prepared with a 50 g dm^{-3} urease solution (you may call this immobilised urease)
- an unlimited volume of 50 g dm^{-3} urease solution (you may call this free urease)
- an unlimited volume of 1.0 mol dm^{-3} urea solution
- an unlimited volume of distilled water
- beakers and flasks of different sizes
- stop watch or electronic timer
- broad and narrow range pH papers and liquids with appropriate colour charts, pH probes and meters
- colorimeter and tubes/cuvettes
- thermometer
- thermostatically-controlled water baths
- graduated pipettes and pipette fillers
- filter funnels
- syringes
- glass rods for stirring
- test-tubes and boiling tubes
- test-tube racks

Your plan should include a clear statement of the hypothesis or prediction and should identify key variables.

Your plan should be written in clear scientific language and should be illustrated with a diagram.

Your plan should give full details and explanations of the procedures that you would adopt to ensure that the results are as precise and reliable as possible. A brief risk assessment should be included.

[Total: 12]

Hypothesis or prediction ;
e.g. rate of hydrolysis is faster using free enzyme/quantity of urea hydrolysed over time is greater with free enzyme/immobilised urease catalyses reaction over much longer period of time

Theory to support candidate's hypothesis or prediction ;
e.g. refs to accessible active sites/diffusion of substrate into alginate beads/stability of enzyme in alginate beads

Outline of strategy ;
e.g. method of following the reaction taking samples at intervals and calculating the initial rate

Justification/evaluation, of strategy ;
e.g. can only alter concentration of immobilised enzyme by changing number of beads/limitations of colour comparison these could be awarded at the end of the plan

Method of determining, pH / (the concentration) of ammonium carbonate, at intervals ;
e.g. use of pH indicator, to follow colour change

At least two control variables ;
e.g. temperature, concentration of urea solution, volumes used, number of beads

Risk assessment ;
ref to hazard and precaution

use range of concentrations of urea ;

.. use range of concentrations of urease ;
to find suitable concentrations to make comparison ..

.. dilution table(s) included ; A ratios ..

.. method to ensure concentration of urease in reaction mixtures is the same for both free and immobilised enzyme ; ..

.. urea solution mixed with pH indicator ; ..

.. equilibration in water bath ; ..

.. mixing, urease/beads, and urea solution at time = 0 ; ..

.. staggered start ; ..

.. samples taken at stated intervals ; ..

.. uncertainty/precision, of results ; ..

.. plot results and take gradient to give initial rate ; ..

.. colour standard set up at known pH ; ..

.. time taken (t) to reach colour standard recorded ; ..

.. rate = $1/t$; A $1000/t$, etc. ..

.. colour change followed in colorimeter ; ..

.. repeats/replicates (calculate means) ; ..

.. calculate, standard deviation/standard error ; ..

.. ref to use of t-test to see if rates are significantly different ; ..

.. plot results on appropriate graph (bar or line) ; ..

.. **12max** ..

[illegible]

[illegible]

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5.

(a) Describe the genetic diseases SCID (severe combined immunodeficiency) and cystic fibrosis. [12]

(b) Discuss the factors which prevent gene therapy from becoming an effective treatment. [8]

(a) describe SCID (severe combined immunodeficiency) and cystic fibrosis. [12]

[Total: 20]

SCID

- S1 more males affected (than females);
- S2 recessive;
- S3 (most common form is) sex-linked (X chromosome);
- S4 mutation of IL2RG (interleukin-2 receptor gamma) gene (for gamma chain on lymphocyte receptor);
- S5 ADA (adenosine deaminase) (deficiency);
- S6 mutation of gene on chromosome 20;
- S7 prevents formation of, lymphocytes / T cells / B cells;
- S8 toxic metabolites not broken down so lymphocytes killed;
- S9 so susceptible to opportunistic infections;
- S10 children usually die (if not treated);
- S11 pneumonia / PCP / meningitis / chickenpox / measles / other valid;

CF

- C1 autosomal/chromosome 7;
- C2 recessive;
- C3 mutation of CFTR/CFTCR gene;
- C4 deletion of 3 base pair/codons;
- C5 CFTR/CFTCR, (protein) has lost phenylalanine;
- C6 (channels) no longer able to transport chloride ions
- C7 causing thick mucus
- C8 resulted in stated problems in lungs/pancreas/reproductive system
- C9 reduces life expectancy

(b) discuss the factors which prevent gene therapy from becoming an effective treatment. [8]

1. difficult to get DNA to integrate into the target cell genome;
2. results can be short-lived / patients need to be treated on a frequent basis;
3. ref. problem with controlling the activity of gene expression;
4. risk of stimulating immune response;
5. virus vector may, regain / develop, virulence;
6. incorrect insertion of gene may cause, named / described, problem to patient;
7. possible problem with finding vector for large gene;
8. many genetic diseases are a result of presence of many defective genes;
9. impossible to introduce many genes at same time;
10. problem dealing with, dominant condition / non dividing cells;
11. ref. problem with liposomes; (toxicity / low efficiency)